

DRCPT-PG: Precision Repair and Proof Review

Source anchor: Rebased_99afee22

19 September 2026

Authorized scope. Resolve I1–I10 in the package-root proof audit and skeleton. Preserve unrelated content. The full original package, including data and historical audit evidence, is retained.

Assurance. Proof-checker and proof-orchestrator were used with fresh source reviews. Reviews are same-family and provisional. The scoped repair result is distinct from unrestricted full-paper acceptance; preexisting reservations are recorded separately in the current PROOF_AUDIT.md and the raw final review.

Issue overview

ID	Severity	Issue	Repair strategy
I1	MINOR	Undefined survival and score shorthands	define existing object
I2	MINOR	Population gradient used before definition	define existing object
I3	MINOR	Trajectory score and population gradient shared g	rename alias
I4	MINOR	Two call-count representations	unify notation
I5	MINOR	Consecutive episode boundary was implicit	explicit interface
I6	MAJOR	Environment model confused with induced law	clarify model interface
I7	MAJOR	Simulator regularity was only implicit	strengthen assumption
I8	MINOR	Unused B_J bound alias	remove unused alias
I9	MAJOR	Global table obscured the semantic chain	simplify table
I10	MINOR	Residual aggregations had ambiguous names	unify terminology

I1: Undefined survival and score shorthands

Location: sections/06_main_results.tex

Before: The gradient identity used undefined $S_{\{k,\theta\}}$ and unidentified G_{θ} .

After: Define S_{θ} directly from $S_{\{P_k, z_k, r_k, \theta\}}$, and G_{θ} from $G_{\{k, \theta\}}$.

Scope / downstream effect: No objective, score, or integral is changed.

I2: Population gradient used before definition

Location: sections/06_main_results.tex

Before: The full/half lemma used g before identifying its fixed target.

After: Define $g := \nabla J(\theta)$ in the lemma before the expansion.

Scope / downstream effect: The existing cancellation proof and constants are unchanged.

I3: Trajectory score and population gradient shared g

Location: appendix/C_estimator.tex

Before: A trajectory realization used g_x and then renamed it G_x .

After: Write $x = (v_+, v_-, G_x)$ directly.

Scope / downstream effect: No random variable or estimator is replaced.

I4: Two call-count representations

Location: *appendix/E_minimax.tex*

Before: The proof changed T_{call} to a $\text{mathsf{var}}$ variant.

After: Use T_{call} consistently; identify its oracle-filtration stopping property.

Scope / downstream effect: The stopped KL argument and expected-call budget remain identical.

I5: Consecutive episode boundary was implicit

Location: *sections/03_problem_setup.tex*

Before: Algorithm 2 returned the next episode state without an explicit date bridge.

After: State $t_{k+1} = t_k + h$ for consecutive episodes.

Scope / downstream effect: No horizon, wealth recurrence, or experiment is changed.

I6: Environment model confused with induced law

Location: *sections/03_problem_setup.tex*

Before: P was called a trajectory law while P_{theta} was immediately used.

After: Identify P as the frozen episode model and P_{theta} as the policy-induced augmented trajectory law.

Scope / downstream effect: Likelihood derivatives act on the policy; no extra persistent model notation is introduced.

I7: Simulator regularity was only implicit

Location: *sections/04_assumptions.tex*

Before: The estimator and online proofs applied standing likelihood bounds under the frozen simulator without explicit quantification.

After: Require the same bounds and parameter-independence conditions uniformly for every analyzed true and frozen-simulator model, conditional on the frozen context.

Scope / downstream effect: This is an explicit assumption-scope strengthening/clarification. It does not certify arbitrary simulators or unlimited-run bounded wealth.

I8: Unused B_J bound alias

Location: *sections/06_main_results.tex*

Before: B_J was defined but the theorem used the already established B_v .

After: Delete only the B_J sentence.

Scope / downstream effect: The endpoint term remains $2 B_v$.

I9: Global table obscured the semantic chain

Location: *appendix/A_notation.tex*

Before: Two tables mixed proof-local constants with core notation and omitted the information/context bridge.

After: Replace them with one 16-row core table; retain all local estimator definitions and explicitly retain market state m_u .

Scope / downstream effect: The final check caught and repaired the temporary loss of the m_u explanation in this table. No analytic lemma below the table is edited.

I10: Residual aggregations had ambiguous names

Location: *setup/results/experiments; Appendix D; Introduction*

Before: Q , its square, its average expectation, and DLR were inconsistently named.

After: Use projected residual, squared projected residual, average expected squared projected residual, and dynamic local regret (DLR).

Scope / downstream effect: Only names are changed; all metric definitions and experimental numerals remain intact. E3 A/B titles now include squared.

The substantive proof obligation: I7

The target is the conditional simulator-gradient identity and the estimator application used by the online theorem. It suffices to establish a theta-independent frozen nonpolicy kernel, a common likelihood-derivative envelope, and the finite moments used by the unchanged LOO proof.

Compact policy parameters and bounded features place all Dirichlet concentrations in a positive compact interval. On a compact neighborhood, a common density/derivative envelope is proportional to

$$\prod_{i=0}^N \tilde{\omega}_i^{a_- - 1} \left(1 + \sum_{i=0}^N |\log \tilde{\omega}_i|^2 \right), \quad a_- > 0.$$

It is integrable over the simplex. A finite product handles the horizon; theta-independent market kernels integrate to one conditionally. DCT therefore yields the likelihood identities under each analyzed frozen simulator. Bounded weight derivatives and a finite integration interval permit CPT differentiation, and Fubini is justified by finite score first moment.

The existing LOO cancellation and telescoping results then give, for the supplied replication schedule,

$$\mathbb{E}[\hat{g}_k^{\text{db}} - \tilde{g}_k(\theta_k) \mid \mathcal{F}_k] = 0, \quad \mathbb{E}[\|\hat{g}_k^{\text{db}} - \tilde{g}_k(\theta_k)\|^2 \mid \mathcal{F}_k] \leq C_{\text{var}}(n)/M_k.$$

The cross term with the frozen simulator-error vector vanishes. The unchanged projected-ascent and moving-objective telescope recover the original online inequality, with the same constants 8, 10 and $2B_v$. Thus the scope repair supplies the premise already required at the actual application site.

Why the scope matters. If a simulator were allowed a theta-dependent Bernoulli return kernel while policy features were zero, the recorded policy score would be zero but the simulator CPT objective could have nonzero derivative. The added parameter-independence condition excludes that construction. This is a verified illustration of the missing condition, not a counterexample to the repaired theorem.

Proof-obligation delta and preserved content

Only the true/simulator scope of the standing regularity hypothesis is made stronger explicitly. The score identity, expansion, telescoping, variance, cost, CLT, projected ascent, DLR and minimax equations retain their values and mathematical content. The error-bound and minimax-oracle assumptions remain separate optional/specialized conditions. No new result or experiment has been claimed.

All original labeled display equations were compared mechanically, allowing only the I4 call-count typography substitution. Experimental numerals, algorithm source, bibliography, manuscript preamble and discussion placeholder were unchanged. Internal references resolve, with no duplicate labels. The final manuscript compiles to 38 pages.

The E3 PDF was edited only in two title text commands. At 108 dpi all changed pixels lie in title rows 10–28; every pixel below row 40 is identical to the source PDF. The source package’s stale two-panel PNG/SVG exports were synchronized from the active four-panel PDF. No plotting data or confidence interval was recalculated for delivery. The plot script changes only the corresponding two title strings.

Evidence and remaining reservations

See the current PROOF_AUDIT.md for the separate scoped/full-manuscript verdicts and the final review in prompts/26091901-01/. Earlier root and nested audits remain historical evidence. The current file hashes and deterministic checks identify exactly what was reviewed. Newly identified preexisting issues outside I1–I10 remain separate; they are not concealed by a scoped repair pass.

This report does not claim cross-family acceptance, unconditional simulator correctness, a stronger asymptotic regime, completed E5–E7 results, or a completed Discussion.